

Multi Skill Technician Electrical

QP Code: ELE/Q3109

Version: 1.0

NSQF Level: 4

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ELE/Q3109: Multi Skill Technician Electrical

Brief Job Description

A Multi Skilled Technician (Electrical) is responsible for finding out and repairing faults in multiple small domestic Home Appliances. The individual installs the appliance and interacts with customers to diagnose the problem and possible causes, rectifies minor problems or replaces faulty modules for failed parts or recommends factory repairs.

Personal Attributes

The job requires the individual to be self-motivated, Inquisitive and analytical. The individual must be willing to work in the field and travel through the day from one customers premise to another. Punctuality, amenable behaviour, patience, good interpersonal relationship building, trustworthiness, integrity and critical thinking are important attributes for this job.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. ELE/N3101: Engage with customer for service
2. ELE/N3118: Install the water purifier
3. ELE/N3119: Repair dysfunctional water purifier
4. ELE/N3120: Repair dysfunctional mixer/juicer/grinder
5. ELE/N3124: Repair small home appliances (Geyser & Fan)
6. ELE/N9302: Diagnose and repair fault in LED Light
7. ELE/N9919: Work with superiors and colleagues
8. ELE/N9921: Follow safety standards
9. FIC/N9005: Evaluate and develop entrepreneur skills

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Consumer Electronics & IT Hardware
Occupation	Sales And After Sales Service

Country	India
NSQF Level	4
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3113.0101
Minimum Educational Qualification & Experience	8th Class
Minimum Level of Education for Training in School	8th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	19/04/2017
Next Review Date	31/03/2020
Deactivation Date	31/03/2020
NSQC Approval Date	
Version	1.0

ELE/N3101: Engage with customer for service

Description

This OS unit is about interacting with customer to understand their requirement with respect to problem in the appliance

Scope

This unit/ task covers the following:

- Interact with the customer prior to visit
- Interact with customer at their premises
- Suggest possible solutions to customer
- Achieve productivity and quality as per company's norms

Elements and Performance Criteria

Interacting with customer prior to visit

To be competent, the user/individual on the job must be able to:

PC1. check customer complaint registered at customer care or installation schedule

PC2. call customer to confirm problem and fix time for visit

PC3. greet the customer and confirm the problem registered

PC4. be polite and patient when interacting with customer

PC5. check about warranty status of appliance and annual maintenance contract

PC6. anticipate possible problems to carry tools and parts accordingly

PC7. ascertain customer location in order to make the route plan for the day

Interacting with customer at their premises

To be competent, the user/individual on the job must be able to:

PC8. enquire about the symptoms and history of problems in the appliance

PC9. ask about the age of appliance and status of upkeep

PC10. identify the problem based on customers information

PC11. communicate the problems identified and educate on possible reasons

PC12. inform about costs involved

Suggesting possible solutions to customer

To be competent, the user/individual on the job must be able to:

PC13. discuss the problem(s) identified with customer

PC14. suggest possible solutions and costs involved

PC15. explain the time required and methodology for servicing necessary

PC16. seek customers approval on further action

Achieving productivity and quality

To be competent, the user/individual on the job must be able to:

PC17. accurately assess the problem and solution(s) necessary

PC18. offer most appropriate and cost-effective service as per customers requirement

PC19. communicate problem effectively in order to secure customers confidence
PC20. ensure customer satisfaction and positive feedback
PC21. record minimum customer complaints post service
PC22. avoid repeat problem post service
PC23. prepare most optimum route plan to complete daily target visits

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. companys policies on: customer care
KU2. companys code of conduct
KU3. organisation culture and typical customer profile
KU4. companys reporting structure
KU5. companys documentation policy
KU6. companys products and recurring problems reported in consumer appliances
KU7. how to communicate with customers in order to put them at ease
KU8. basic electrical and mechanical modules of various appliances
KU9. electronics involved in the type of appliance
KU10. models of different appliances and their common and distinguishing features
KU11. functionality of different features of appliances and new features
KU12. etiquette to be followed at customers premises
KU13. precautions to be taken while handling field calls and dealing with customers
KU14. relevant reference sheets, manuals and documents to carry in the field

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. how to read product and module serial numbers and interpret details such as make, date, availability
GS2. how to note problems on job sheet and details of work done
GS3. to put customer at ease and generate customers confidence
GS4. to listen carefully and interpret their statement of symptoms
GS5. to seek inputs at assess the problems
GS6. how to communicate in local language
GS7. how to educate and inform customer about contractual issues such as warranty, cost of service and module replacement
GS8. to educate on precautions to be taken post repairs to avoid recurrence of problem
GS9. importance of personal grooming
GS10. significance of etiquette such as maintaining the appropriate physical distance with customer during conversation, not entering bedroom without permission
GS11. importance of being patient and courteous with all types of customers

GS12. being polite and courteous under all circumstances

GS13. importance of maintaining clean surface/work area

GS14. decide on the spot on whether interaction of customer with supervisor is necessary or not

GS15. when to call customer care and close the call after work is done to customers satisfaction and documentation is complete

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Interacting with customer prior to visit	**7**	**14**	**-**	**-**
PC1. check customer complaint registered at customer care or installation schedule	1	2	-	-
PC2. call customer to confirm problem and fix time for visit	1	2	-	-
PC3. greet the customer and confirm the problem registered	1	2	-	-
PC4. be polite and patient when interacting with customer	1	2	-	-
PC5. check about warranty status of appliance and annual maintenance contract	1	2	-	-
PC6. anticipate possible problems to carry tools and parts accordingly	1	2	-	-
PC7. ascertain customer location in order to make the route plan for the day	1	2	-	-
Interacting with customer at their premises	**10**	**14**	**-**	**-**
PC8. enquire about the symptoms and history of problems in the appliance	2	2	-	-
PC9. ask about the age of appliance and status of upkeep	2	3	-	-
PC10. identify the problem based on customers information	2	3	-	-
PC11. communicate the problems identified and educate on possible reasons	2	3	-	-
PC12. inform about costs involved	2	3	-	-
Suggesting possible solutions to customer	**8**	**12**	**-**	**-**
PC13. discuss the problem(s) identified with customer	2	3	-	-
PC14. suggest possible solutions and costs involved	2	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. explain the time required and methodology for servicing necessary	2	3	-	-
PC16. seek customers approval on further action	2	3	-	-
Achieving productivity and quality	**15**	**20**	**-**	**-**
PC17. accurately assess the problem and solution(s) necessary	2	3	-	-
PC18. offer most appropriate and cost-effective service as per customers requirement	2	3	-	-
PC19. communicate problem effectively in order to secure customers confidence	2	3	-	-
PC20. ensure customer satisfaction and positive feedback	2	3	-	-
PC21. record minimum customer complaints post service	2	3	-	-
PC22. avoid repeat problem post service	2	3	-	-
PC23. prepare most optimum route plan to complete daily target visits	3	2	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N3101
NOS Name	Engage with customer for service
Sector	Electronics
Sub-Sector	Consumer Electronics & IT Hardware
Occupation	AFTER SALES SERVICE
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/12/2013
Next Review Date	31/03/2022
NSQC Clearance Date	NA

ELE/N3118: Install the water purifier

Description

This OS unit is about installing the newly purchased water purifier at customers location and make it ready to use.

Scope

This unit/ task covers the following:

- Undertake pre-installation site visit
- Remove packaging and check accessories
- Fix the water purifier at identified location
- Check water purifier's functioning
- Complete the documentation
- Interact with supervisor or superior
- Achieve productivity and quality as per company's norms

Elements and Performance Criteria

Undertake pre-installation site visit

To be competent, the user/individual on the job must be able to:

PC1.. visit the customers premise before carrying out the installation

PC2.. interact with the customer to understand whether the water purifier would be placed under the sink (UTS) or on the wall

PC3.. check that the location meets structural requirements such as distance from power supply, vicinity to plumbing point, etc.

PC4.. make the customer aware of any pre-installations/masonry/electrical work to be carried out and educate the customer about requirement of adequate water pressure at the inlet source

PC5.. make necessary markings for placement of the water purifier unit

PC6.. seek appointment for the next visit

PC1.. visit the customer's premise before carrying out the installation

Remove packaging and check accessories

To be competent, the user/individual on the job must be able to:

PC7.. remove the packaging in which the purifier was shipped to customer from point of sale/ warehouse

PC8.. check that the product matches the customer order in terms of colour and make

PC9.. check that all supporting accessories purchased have are there in the pack

PC10. check that tools and fitments required for the installation are available

PC11. clear up the packaging material waste and dispose as per companys norms

PC11. clear up the packaging material waste and dispose as per company's norms

Fix the water purifier at identified location

To be competent, the user/individual on the job must be able to:

PC12. check if pre-installation requirements are met

PC13. make measurements at the location identified and drill holes ensuring no internal wiring damage takes place

PC14. mount the filter and ensure that the screws are fastened securely

PC15. drain the inlet line before connecting it to the water purifier

PC16. connect the outlet pipe to the drain (if applicable)

PC17. connect the purifier to the nearest power supply point

Check waterpurifier's functioning

To be competent, the user/individual on the job must be able to:

PC18. ensure that the filter is aligned as per instructions in the installation manual

PC19. run the purifier and ensure there are no leaks at any point

PC20. demonstrate the features and utility to the customer

PC21. explain maintenance procedures to be followed while using the water purifier

Complete the documentation

To be competent, the user/individual on the job must be able to:

PC22. fill in customer acknowledgement form

PC23. seek customers signature

PC24. complete other documentation for recording completion of installation

PC25. call customer care and inform about job completed

PC23. seek customer's signature

Interact with supervisor or superior

To be competent, the user/individual on the job must be able to:

PC26. understand the work requirement from superior, periodically

PC27. report to superior on the work completed

PC28. escalate the customer issues and problems that are unresolved in the field

PC29. document the work completed on the company ERP software for tracking and future references

Achieve productivity and quality as per company's norms

To be competent, the user/individual on the job must be able to:

PC30. remove packaging without damage to the water purifier unit and accessories

PC31. position the water filter as per requirements specified in instructions manual

PC32. educate customer on importance of proper placing

PC33. carry and use the correct tools and equipment for installation

PC34. operate and check that they are in a safe and stable condition

PC35. complete installation in time target given

PC36. educate customer on proper operation and maintenance procedures

PC37. complete daily field schedule as per instructions/format within the designated time

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. companys policies on: incentives, delivery standards, and personnel management, call closure

KU2. sales, installation and after sales support policy
KU3. importance of the individuals role in the workflow
KU4. reporting structure
KU5. companys policy on products warranty and other terms and conditions
KU6. installation site requirements (structural requirements, plumbing etc.)
KU7. water flow diagram and electrical circuit diagram of the Water Purifier
KU8. water purification process and different layers of filter present within the unit such as sediment filter, carbon block filter, TFC/TFM membrane, inline carbon filter etc
KU9. different technologies in water purification (such as reverse osmosis, deionisation etc.)
KU10. different features and functionalities of various models
KU11. safety precautions to be taken while installing
KU12. manual-based procedure of installing the water purifier
KU13. packaging waste disposal procedures
KU14. other products of the company
KU15. safety rules, policies and procedures
KU16. quality standards to be followed

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. to read job sheet for installation as registered by customer care/companys ERP system
GS2. to document the completed work
GS3. to read the standard operating procedures for different types of water purifiers
GS4. to read about different types of water purifiers, their basic electro-mechanical structure and functionality of features
GS5. to share work load as required
GS6. to achieve the targets given on installation per day or month
GS7. the water flows through various switches, valves and different layers of filter
GS8. to operate the water purifier and use the various features
GS9. to fix various accessories and parts that have accompanied the unit
GS10. to check features and functionalities after installation
GS11. to operate tools such as electric drill, screw drivers, wrenches, tube cutters/benders, spanners, etc.
GS12. to make appropriate settings after plugging in
GS13. to improve work processes
GS14. to reduce repetition of errors in installation
GS15. to spot process disruptions and delays
GS16. to report on any customer concerns to superiors without delay

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Undertake pre-installation site visit	**6**	**7**	**-**	**-**
PC1.. visit the customers premise before carrying out the installation	1	1	-	-
PC2.. interact with the customer to understand whether the water purifier would be placed under the sink (UTS) or on the wall	1	2	-	-
PC3.. check that the location meets structural requirements such as distance from power supply, vicinity to plumbing point, etc.	1	1	-	-
PC4.. make the customer aware of any pre-installations/masonry/electrical work to be carried out and educate the customer about requirement of adequate water pressure at the inlet source	1	1	-	-
PC5.. make necessary markings for placement of the water purifier unit	1	1	-	-
PC6.. seek appointment for the next visit	1	1	-	-
PC1.. visit the customer's premise before carrying out the installation	1	1	-	-
Remove packaging and check accessories	**5**	**5**	**-**	**-**
PC7.. remove the packaging in which the purifier was shipped to customer from point of sale/ warehouse	1	1	-	-
PC8.. check that the product matches the customer order in terms of colour and make	1	1	-	-
PC9.. check that all supporting accessories purchased have are there in the pack	1	1	-	-
PC10. check that tools and fitments required for the installation are available	1	1	-	-
PC11. clear up the packaging material waste and dispose as per companys norms	1	1	-	-
PC11. clear up the packaging material waste and dispose as per company's norms	1	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Fix the water purifier at identified location	**7**	**12**	**-**	**-**
PC12. check if pre-installation requirements are met	1	2	-	-
PC13. make measurements at the location identified and drill holes ensuring no internal wiring damage takes place	2	2	-	-
PC14. mount the filter and ensure that the screws are fastened securely	1	2	-	-
PC15. drain the inlet line before connecting it to the water purifier	1	2	-	-
PC16. connect the outlet pipe to the drain (if applicable)	1	2	-	-
PC17. connect the purifier to the nearest power supply point	1	2	-	-
Check waterpurifier's functioning	**7**	**13**	**-**	**-**
PC18. ensure that the filter is aligned as per instructions in the installation manual	2	3	-	-
PC19. run the purifier and ensure there are no leaks at any point	2	3	-	-
PC20. demonstrate the features and utility to the customer	1	4	-	-
PC21. explain maintenance procedures to be followed while using the water purifier	2	3	-	-
Complete the documentation	**4**	**6**	**-**	**-**
PC22. fill in customer acknowledgement form	1	1	-	-
PC23. seek customers signature	1	1	-	-
PC24. complete other documentation for recording completion of installation	1	2	-	-
PC25. call customer care and inform about job completed	1	2	-	-
PC23. seek customer's signature	1	1	-	-
Interact with supervisor or superior	**4**	**8**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. understand the work requirement from superior, periodically	1	2	-	-
PC27. report to superior on the work completed	1	2	-	-
PC28. escalate the customer issues and problems that are unresolved in the field	1	2	-	-
PC29. document the work completed on the company ERP software for tracking and future references	1	2	-	-
Achieve productivity and quality as per company's norms	**7**	**9**	**-**	**-**
PC30. remove packaging without damage to the water purifier unit and accessories	1	1	-	-
PC31. position the water filter as per requirements specified in instructions manual	1	1	-	-
PC32. educate customer on importance of proper placing	1	1	-	-
PC33. carry and use the correct tools and equipment for installation	1	1	-	-
PC34. operate and check that they are in a safe and stable condition	1	1	-	-
PC35. complete installation in time target given	-	2	-	-
PC36. educate customer on proper operation and maintenance procedures	1	1	-	-
PC37. complete daily field schedule as per instructions/format within the designated time	1	1	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N3118
NOS Name	Install the water purifier
Sector	Electronics
Sub-Sector	Consumer Electronics & IT Hardware
Occupation	After Sales Service
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/03/2014
Next Review Date	31/03/2022
NSQC Clearance Date	NA

ELE/N3119: Repair dysfunctional water purifier

Description

This OS unit is about understanding the customers complaints, identifying the fault and fixing the water purifier.

Scope

This unit/ task covers the following:

- Understand the symptoms in the water purifier and identify the fault
- Replace dysfunctional part in the water purifier unit
- Confirm functionality of the repaired unit
- Achieve productivity and quality as per company's norms

Elements and Performance Criteria

Understand the symptoms in the water purifier and identify the fault

To be competent, the user/individual on the job must be able to:

PC1.. diagnose the fault based on customer interaction and initial inspection

PC2.. check if the water pressure is as specified by company standards

PC3.. shut off the system by turning of water supply and unplug the unit

PC4.. place a piece of cloth or towel under the unit in order to avoid any water spills on the floor

PC5.. carry out basic inspection of feed water valve, tank valve, tubing, housing etc.

PC6.. separate and inspect every part of the unit if the fault is not identified through basic inspection

PC7.. send to factory for in depth diagnosis, if problem remains un-identified at site

Replace dysfunctional part in water purifier unit

To be competent, the user/individual on the job must be able to:

PC8.. replace component at location, if the fault identified is because of damage of components such as valves or wearing out of membrane or filter

PC9.. remove and replace the faulty module with a functional one, either on a second visit or as pre-identified and collected from the service centre, if the problem is at the PCB level or components that cannot be replaced at site

Confirm functionality of the repaired unit

To be competent, the user/individual on the job must be able to:

PC10. reassemble the unit

PC11. start supply of water to the unit and confirm that unit is functioning

PC12. check that all the modules of the unit work as per specifications

PC13. demonstrate and confirm functionality of the unit with customer

PC14. educate the customer about cleaning procedures and other best practices

PC15. collect necessary payments from the customer, if applicable

PC16. fill in customer acknowledgement form

PC17. complete other documentation procedures to record complaint closure

Achieve productivity and quality as per company's norms

To be competent, the user/individual on the job must be able to:

PC18. ensure damage free handling of the unit

PC19. diagnose the problem accurately and in assigned time

PC20. identify the problem modules accurately such as inlet valve, auto shut off valve, saddle valve, housing, O ring, PCB

PC21. fix the dysfunctional water purifier in designated time

PC22. rectify completely to avoid repeat fault in the water purifier

PC23. record minimum customer complaints post service

PC24. meet daily target on attending to number of complaints

PC25. select the right spares according to recorded complaints at the customer care

PC26. clearly communicate type of module required to the service centre, if a faultymodule is to be replaced

PC27. secure repairs completion receipt from customer

PC28. educate customer on water purifier maintenance and correct practices to follow in order to avoid further problems

PC29. ensure 100% customer satisfaction

PC30. recover payments as per rate sheet/ communication from customer care

PC31. sell related products such as new equipment or Annual Maintenance Contracts (AMC) as per company policy

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. companys policies on: incentives, delivery standards and personnel management and customer service standards

KU2. reporting and documentation processes

KU3. water purifier manufacturing capabilities of the organization

KU4. importance of the individuals role in the system

KU5. reporting structure

KU6. water flow diagram and electrical circuit diagram of the Water Purifier

KU7. water purification process and different layers of filter present within the unit such as sediment filter, carbon block filter, TFC/TFM membrane, inline carbon filter etc.

KU8. different technologies in water purification (such as reverse osmosis etc.)

KU9. parameters such as production rate, water chemistry, drain rate, input water pressure/temperature etc.

KU10. different types of water purifiers manufactured by the company

KU11. features of different water purifier models of the company

KU12. functioning of the Water Purifier and its various filters

KU13. basic electronics (knowledge of components such as diode, transformer, LED,photo transistor, capacitor, resistor, inductor, thermistor, Ics

KU14. chemical and other properties of various filters of the Water Purifier

KU15. fundamentals of electricity such as ohms law, difference between ac and dc,calculation of energy consumption of Water Purifiers, understanding of domestic wiring, understanding of series and parallel connections

KU16. troubleshooting knowledge with respect to water purifiers

KU17. hazards, their causes and prevention/personal safety

KU18. frequently occurring faults such as low/no water production, leaks, bad tasting water etc.

KU19. components/modules of the water purifier and their prices

KU20. other products of the company

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. how to read warnings, instructions and other text material on product labels, and components

GS2. how to read job sheet and complaints

GS3. how to read product operating manuals

GS4. how to operate computers and software installed

GS5. how to read and understand electrical and electronic symbols, multiples and SI units

GS6. how to document completion note for customer

GS7. how to record completion information in the ERP system

GS8. to operate/use TDS tester, tube cutter, tube bender, temperature meter, pressure gauges, wrenches, pliers, screw drivers

GS9. to detect basic electrical faults such as improper/no earth, defective power cord, connector or internal wiring defect, short/ loose/open contacts, blown fuse

GS10. company specified procedures to change filters, resin and membrane of different models of water purifier

GS11. to diagnose reasons for low/no water production due to feed water valve or tank valve not being on or kinked tubing

GS12. to identify reasons for leaks in the filter housing due to loose housing, damaged or misaligned O ring, cracks in the housing

GS13. to detect worn out auto shut off valve through symptoms such as loud vibrating noise, drain water never shutting off etc.

GS14. to detect other problems such as clogged filters, storage tank problems, clogged flow resistor, inadequate/excessive water pressure, improper saddle valve mounting etc.

GS15. how to interact with customer to understand the problem faced

GS16. how to market and sell accessories and products of the company

GS17. importance of communicating in language

GS18. Precautions and etiquette while dealing with customer

GS19. be polite, patient and punctual

GS20. to match symptoms of the fault noticed to the cause of the problem

GS21. anticipate and avoid hazards that may occur during repairs because of tools, materials used or repair processes

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Understand the symptoms in the water purifier and identify the fault	**10**	**14**	**-**	**-**
PC1.. diagnose the fault based on customer interaction and initial inspection	2	2	-	-
PC2.. check if the water pressure is as specified by company standards	1	2	-	-
PC3.. shut off the system by turning of water supply and unplug the unit	1	2	-	-
PC4.. place a piece of cloth or towel under the unit in order to avoid any water spills on the floor	1	2	-	-
PC5.. carry out basic inspection of feed water valve, tank valve, tubing, housing etc.	1	2	-	-
PC6.. separate and inspect every part of the unit if the fault is not identified through basic inspection	2	2	-	-
PC7.. send to factory for in depth diagnosis, if problem remains un-identified at site	2	2	-	-
Replace dysfunctional part in water purifier unit	**8**	**14**	**-**	**-**
PC8.. replace component at location, if the fault identified is because of damage of components such as valves or wearing out of membrane or filter	4	7	-	-
PC9.. remove and replace the faulty module with a functional one, either on a second visit or as pre-identified and collected from the service centre, if the problem is at the PCB level or components that cannot be replaced at site	4	7	-	-
Confirm functionality of the repaired unit	**10**	**16**	**-**	**-**
PC10. reassemble the unit	1	2	-	-
PC11. start supply of water to the unit and confirm that unit is functioning	1	2	-	-
PC12. check that all the modules of the unit work as per specifications	2	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. demonstrate and confirm functionality of the unit with customer	2	2	-	-
PC14. educate the customer about cleaning procedures and other best practices	1	2	-	-
PC15. collect necessary payments from the customer, if applicable	1	2	-	-
PC16. fill in customer acknowledgement form	1	2	-	-
PC17. complete other documentation procedures to record complaint closure	1	2	-	-
Achieve productivity and quality as per company's norms	**12**	**16**	**-**	**-**
PC18. ensure damage free handling of the unit	1	1	-	-
PC19. diagnose the problem accurately and in assigned time	1	1	-	-
PC20. identify the problem modules accurately such as inlet valve, auto shut off valve, saddle valve, housing, O ring, PCB	1	1	-	-
PC21. fix the dysfunctional water purifier in designated time	1	1	-	-
PC22. rectify completely to avoid repeat fault in the water purifier	1	1	-	-
PC23. record minimum customer complaints post service	1	1	-	-
PC24. meet daily target on attending to number of complaints	1	1	-	-
PC25. select the right spares according to recorded complaints at the customer care	1	1	-	-
PC26. clearly communicate type of module required to the service centre, if a faultymodule is to be replaced	1	1	-	-
PC27. secure repairs completion receipt from customer	1	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. educate customer on water purifier maintenance and correct practices to follow in order to avoid further problems	-	2	-	-
PC29. ensure 100% customer satisfaction	-	2	-	-
PC30. recover payments as per rate sheet/ communication from customer care	1	1	-	-
PC31. sell related products such as new equipment or Annual Maintenance Contracts (AMC) as per company policy	1	1	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N3119
NOS Name	Repair dysfunctional water purifier
Sector	Electronics
Sub-Sector	Consumer Electronics & IT Hardware
Occupation	After Sales Service
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/03/2014
Next Review Date	31/03/2022
NSQC Clearance Date	NA

ELE/N3120: Repair dysfunctional mixer/juicer/grinder

Description

This OS unit is about understanding the customers complaints, identifying the fault and fixing the mixer/juicer/grinder.

Scope

This unit/ task covers the following:

- Understand the symptoms in the appliance and identify the fault
- Replace dysfunctional part of the small appliance
- Confirm functionality of the repaired unit
- Achieve productivity and quality as per company's norms

Elements and Performance Criteria

Understand the symptoms in the appliance and identify the fault

To be competent, the user/individual on the job must be able to:

PC1. understand usage pattern of the mixer/grinder from the customer

PC2. diagnose the fault based on customer interaction and initial inspection

PC3. unplug the unit, turn overload switch back to original position if the appliance turned off due to overload

PC4. carry out basic tests such as power supply inspection, volt ampere test and earth test power supply

PC5. separate and inspect every module of the unit if the fault is not identified through basic tests

PC6. send to factory for in depth diagnosis, if problem remains un-identified at site

Replace dysfunctional part of the small appliance

To be competent, the user/individual on the job must be able to:

PC7. replace component at location, if the fault identified is because of damage of components such as relay or thermostat

PC8. remove and replace the faulty module with a functional one, either on a second visit or as pre-identified and collected from the service centre, if the problem is at the PCB level or components that cannot be replaced at site

Confirm functionality of the repaired unit

To be competent, the user/individual on the job must be able to:

PC9. reassemble the unit

PC10. switch on power supply and confirm that unit is functioning

PC11. demonstrate and confirm functionality of the unit with customer

PC12. educate the customer about cleaning procedures, using different jars for different purposes and other best practices

PC13. collect necessary payments from the customer, if applicable

PC14. fill in customer acknowledgement form

PC15. complete other documentation procedures to record complaint closure

Achieve productivity and quality as per companys norms

To be competent, the user/individual on the job must be able to:

PC16. ensure damage free handling of the unit

PC17. diagnose the problem accurately and in assigned time

PC18. identify the problem modules accurately such as the power supply, overload circuit breaker, motors, PCB

PC19. fix the dysfunctional appliance in designated time

PC20. rectify completely to avoid repeat fault in the appliance

PC21. record minimum customer complaints post service

PC22. meet daily target on attending to number of complaints

PC23. select the right spares according to recorded complaints at the customer care

PC24. clearly communicate type of module required to the service centre, if a faulty module is to be replaced

PC25. secure repairs completion receipt from customer

PC26. educate customer on maintenance and correct practices to follow in order to avoid further problems

PC27. ensure 100% customer satisfaction

PC28. recover payments as per rate sheet/ communication from customer care

PC29. sell related products such as new equipment or Annual Maintenance Contracts (AMC) as per company policy

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. companys policies on: incentives, delivery standards and personnel management and customer service standards

KU2. reporting and documentation processes

KU3. importance of the individuals role in the system

KU4. reporting structure

KU5. different models of mixers, grinders and their features and functionalities

KU6. functioning of the appliance and its various modules

KU7. basic electrical fundamentals with regard to functioning of motors, circuit breakers, etc.

KU8. basic electronics (knowledge of components such as diode, transformer, LED, photo transistor, capacitor, resistor, inductor, thermistor, ICs

KU9. functioning of various electromechanical parts of the mixer/grinder

KU10. fundamentals of electricity such as ohms law, difference between ac and dc, calculation of energy consumption of appliances, understanding of domestic wiring, understanding of series and parallel connections

KU11. troubleshooting knowledge with respect to small home appliances

KU12. hazards, their causes and prevention/personal safety

KU13. frequently occurring faults such as abnormal noise during use, jars overflowing, jar leaking etc.

KU14. components/modules of different small home appliances and their prices

KU15. energy ratings such BEE rating and concepts of e waste

KU16. other products of the company

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. how to read warnings, instructions and other text material on product labels, and components

GS2. how to read job sheet and complaints

GS3. how to read product operating manual

GS4. how to read and understand electrical and electronic symbols, multiples and SI units

GS5. how to document completion note for customer

GS6. how to record completion information in the ERP system

GS7. to operate/use multi-meter, clamp meter, tube cutter, tube bender, screw drivers, wrenches, pliers etc.

GS8. to detect basic electrical faults such as improper/no earth, defective power cord, connector or internal wiring defect, short/ loose/open contacts, blown fuse

GS9. to diagnose reasons for abnormal noise during use such as loose jar coupler, overloading of jar, worn out blade shaft, worn out jar bush, worn out/broken motor coupler

GS10. to diagnose reasons for appliance not running due to dysfunctional motor, overload circuit breaker tripping, no power supply etc.

GS11. to identify reasons for overflowing/leaking of contents from the jar such as faulty fitting of dome lid cap, dome gasket, overloading of the jar etc.

GS12. to detect problems in the indicator switch due to lack of power supply, tripping of overload circuit breaker etc.

GS13. how to interact with customer to understand the problem faced

GS14. how to market and sell accessories and products of the company

GS15. importance of communicating in language

GS16. precautions and etiquette while dealing with customer

GS17. be polite, patient and punctual

GS18. to match symptoms of the fault noticed to the cause of the problem

GS19. anticipate and avoid hazards that may occur during repairs because of tools, materials used or repair processes

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Understand the symptoms in the appliance and identify the fault	**12**	**13**	**-**	**-**
PC1. understand usage pattern of the mixer/grinder from the customer	2	2	-	-
PC2. diagnose the fault based on customer interaction and initial inspection	2	3	-	-
PC3. unplug the unit, turn overload switch back to original position if the appliance turned off due to overload	2	2	-	-
PC4. carry out basic tests such as power supply inspection, volt ampere test and earth test power supply	2	2	-	-
PC5. separate and inspect every module of the unit if the fault is not identified through basic tests	2	2	-	-
PC6. send to factory for in depth diagnosis, if problem remains un-identified at site	2	2	-	-
Replace dysfunctional part of the small appliance	**8**	**16**	**-**	**-**
PC7. replace component at location, if the fault identified is because of damage of components such as relay or thermostat	4	8	-	-
PC8. remove and replace the faulty module with a functional one, either on a second visit or as pre-identified and collected from the service centre, if the problem is at the PCB level or components that cannot be replaced at site	4	8	-	-
Confirm functionality of the repaired unit	**10**	**14**	**-**	**-**
PC9. reassemble the unit	2	2	-	-
PC10. switch on power supply and confirm that unit is functioning	1	2	-	-
PC11. demonstrate and confirm functionality of the unit with customer	2	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. educate the customer about cleaning procedures, using different jars for different purposes and other best practices	2	2	-	-
PC13. collect necessary payments from the customer, if applicable	1	2	-	-
PC14. fill in customer acknowledgement form	1	2	-	-
PC15. complete other documentation procedures to record complaint closure	1	2	-	-
Achieve productivity and quality as per companys norms	**10**	**17**	**-**	**-**
PC16. ensure damage free handling of the unit	1	1	-	-
PC17. diagnose the problem accurately and in assigned time	1	1	-	-
PC18. identify the problem modules accurately such as the power supply, overload circuit breaker, motors, PCB	1	1	-	-
PC19. fix the dysfunctional appliance in designated time	1	1	-	-
PC20. rectify completely to avoid repeat fault in the appliance	1	1	-	-
PC21. record minimum customer complaints post service	1	1	-	-
PC22. meet daily target on attending to number of complaints	-	1	-	-
PC23. select the right spares according to recorded complaints at the customer care	1	1	-	-
PC24. clearly communicate type of module required to the service centre, if a faulty module is to be replaced	-	2	-	-
PC25. secure repairs completion receipt from customer	1	1	-	-
PC26. educate customer on maintenance and correct practices to follow in order to avoid further problems	1	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure 100% customer satisfaction	-	2	-	-
PC28. recover payments as per rate sheet/ communication from customer care	-	2	-	-
PC29. sell related products such as new equipment or Annual Maintenance Contracts (AMC) as per company policy	1	1	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N3120
NOS Name	Repair dysfunctional mixer/juicer/grinder
Sector	Electronics
Sub-Sector	Consumer Electronics & IT Hardware
Occupation	After Sales Service
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/03/2014
Next Review Date	31/03/2022
NSQC Clearance Date	NA

ELE/N3124: Repair small home appliances (Geyser & Fan)

Description

This OS unit is about understanding the customers complaints, identifying the fault and fixing the Ceiling Fan / Geyser

Scope

This unit/ task covers the following:

- Understand the symptoms in the Ceiling Fan / Geyser and identify the fault
- Complete the repairs by correcting the faults by replacing the dysfunctional component/module in the Ceiling Fan / Geyser
- Confirm functionality of the repaired unit and post repair activities
- Interact with & train technician
- Achieve productivity & quality as per company's norms
- Ensure safety at workplace

Elements and Performance Criteria

Understand the symptoms in the Ceiling Fan / Geyser and identify the fault

To be competent, the user/individual on the job must be able to:

PC1. understand the complete functioning of the Ceiling Fan / Geyser & manufacturers recommendation(s) on the diagnostic & service procedures to be followed

PC2. understand the possible fault (s) based on the symptoms observed & also reported by customer

PC3. use a range of diagnostic documents like service manual, circuit/wiring diagrams, fault analysis charts, flow charts to investigate the fault

PC4. use a range of tools & equipment to carry out the diagnostic tests to investigate the fault

PC5. perform basic diagnostic tests to confirm that the power source is providing requisite inputs to the Ceiling Fan / Geyser & there are no fault with the same and there are no external faults like improper mounting etc.

PC6. isolate the Ceiling Fan / Geyser / Geyser from the power source & disassemble the component(s) or module(s) from the Ceiling Fan / Geyser and perform the functional test(s) to confirm their working

PC7. inform the superior if unable to follow the company guidelines and identify the fault

Complete the repairs by correcting the faults by replacing the dysfunctional component/module in the Ceiling Fan / Geyser

To be competent, the user/individual on the job must be able to:

PC8. test the basic electrical & electronic components using multi-meter

PC9. check the mains outlet with a multi-meter for correct voltage output

PC10. correct the minor external faults like loose connections, improper mounting

PC11. educate the customer to get the major external faults like faulty or intermittent power

PC12. replace the faulty component or module and re assemble the Ceiling Fan / Geyser

PC13. take help from assistant/junior technician (if any) & train them about the repair process

PC14. follow the companys guidelines in case the replacement component(s) or module(s) are not available or any such scenario

PC15. update the supervisor & the customer in the above situation

Confirm functionality of the repaired unit and post repair activities

To be competent, the user/individual on the job must be able to:

PC16. switch ON the power supply to the Ceiling Fan / Geyser and carry out the functionality test a per the service manual and confirm that it is working normally

PC17. demonstrate and confirm the functionality of the Ceiling Fan / Geyser to the customer

PC18. complete documentation procedures to record complaint closure, collect the applicable charges from the customer & provide receipt of the same as per company policies

PC19. inform the customer about the repairs carried out, preventive care to be taken by them and the usage tips if any and answer to the queries regarding the Ceiling Fan / Geyser or the faults politely to their satisfaction

PC20. offer the recommended accessories & extended warranties/AMC to the customer if available.

PC21. clean the repair area of all the debris & dispose them with customers permission

PC22. collect the defective component(s) or module(s) & handover them to stores

PC23. thank the customer for giving us a chance to serve them & request them to give a positive feedback

PC24. inform the superior about the job closure

Interact with & train technician

To be competent, the user/individual on the job must be able to:

PC25. interact with service technicians from time to time in-order to understand problems faced on the field

PC26. educate junior level technicians about the commonly occurring problems,diagnosis procedures & customer handling

Achieve productivity & quality as per companys norms

To be competent, the user/individual on the job must be able to:

PC27. diagnose the problem accurately and in short time

PC28. identify the problem modules such as the motor, bush, bearing etc

PC29. optimise the time taken to fix the dysfunctional Ceiling Fan / Geyser

PC30. rectify to avoid the repeat fault in the Ceiling Fan / Geyser

PC31. record minimum customer complaints post service

PC32. select the right spares as per recorded complaints at the customer care

PC33. educate customer on Ceiling Fan / Geyser maintenance to avoid problems in the future

PC34. ensure damage free handling of the Ceiling Fan / Geyser

PC35. achieve 100% customer satisfaction

PC36. make sale of related products such as new equipment or AMC as applicable

Ensure safety at workplace

To be competent, the user/individual on the job must be able to:

PC37. comply with health and safety, environmental and other relevant regulations and organizational guidelines at work

PC38. adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while performing the repair operations

PC39. ensure work area is clean and safe from hazards

PC40. ensure that all tools, equipment, power tool cables, extensions are in a safe and usable condition

PC41. dispose of waste items in a safe and environmentally acceptable manner

PC42. leave the work area in a safe condition and free from foreign object debris

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. companys policies on: productivity, incentives, service delivery procedures, quality standard & personnel management and customer service standards relevant to the role

KU2. companys sales and after sales support policy installation, warranty terms, schedule of charges for post warranty jobs, applicable taxes, service documentation

KU3. reporting structure, interdependent functions, lines & procedure in work area including the quality and productivity targets, incentive/penalty structure

KU4. relevant health & safety equipment applicable in the workplace

KU5. Importance of the individuals role in the workflow, responsibilities and sources for information pertaining to employment terms, entitlements, job roles and responsibilities

KU6. escalation matrix and procedures for reporting work and employment related issues

KU7. companys recommended accessories, extended warranty & AMC schemes for the Ceiling Fan / Geyser / Geyser if any

KU8. basic arithmetic calculations & units measurement & multipliers

KU9. KB2. basic knowledge electrical & electronic principles & their applications

KU10. recommended tools & equipment usage/handling instructions & safety precautions including the use of PPE (personal protective equipment) like hand gloves, type of ladder to be used, precautions while using ladder etc.

KU11. the complete functioning of the Ceiling Fan / Geyser, operation sequence & recommended power inputs

KU12. identification of basic electronic/mechanical components or modules in the Ceiling Fan / Geyser, their working & recommended test procedure

KU13. electrical circuit diagram & mechanical assembly & instrumentation set of the Ceiling Fan / Geyser

KU14. recommended installation procedure and its importance in optimum performance of the Ceiling Fan / Geyser

KU15. functionality testing and demonstration of the Ceiling Fan / Geyser

KU16. KB9. working of motor used in the Ceiling Fan / Geyser

KU17. controlling the speed of fan with the regulator

KU18. basic & advance troubleshooting knowledge of the Ceiling Fan / Geyser

KU19. frequently occurring faults of the Ceiling Fan / Geyser & their remedies/fixes

KU20. the recommended procedure of disassembly & re assembly of the Ceiling Fan/ Geyser

KU21. specific safe working practices, breakdown service procedures

KU22. hazards associated with breakdown service procedure & how to minimise them

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. complete forms such as work orders, invoices, maintenance records
GS2. note problems on job sheet and details of work done
GS3. read warnings, instructions and other text material on product labels, components, etc.
GS4. read job sheets or work orders
GS5. read product and module serial numbers and interpret details such as make, date, availability
GS6. receive and ask for clarifications from supervisor on the job requirement
GS7. listen carefully to customer and interpret customers statement of symptoms
GS8. communicate in local language
GS9. educate and inform customer about product, contractual issues such as warranty, cost of service and module replacement
GS10. educate on precautions to be taken in order to avoid recurrence of problem
GS11. follow standard operating procedures while making decisions
GS12. take approval from supervisor in case the decision has to be made for exceptions
GS13. work with supervisor and co-workers to achieve smooth workflow
GS14. work with superiors and co-workers to share knowledge and learning
GS15. maintain personal grooming
GS16. be polite, patient and courteous under all circumstances with all types of customers
GS17. decide on the spot on whether interaction of customer with superior is necessary or not
GS18. maintain proper etiquette including the appropriate physical distance with customer during conversation, not entering bedroom without permission
GS19. put customer at ease and generate customers confidence

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Understand the symptoms in the Ceiling Fan / Geyser and identify the fault	**6**	**12**	**-**	**-**
PC1. understand the complete functioning of the Ceiling Fan / Geyser & manufacturers recommendation(s) on the diagnostic & service procedures to be followed	1	1	-	-
PC2. understand the possible fault (s) based on the symptoms observed & also reported by customer	-	2	-	-
PC3. use a range of diagnostic documents like service manual, circuit/wiring diagrams, fault analysis charts, flow charts to investigate the fault	1	1	-	-
PC4. use a range of tools & equipment to carry out the diagnostic tests to investigate the fault	1	1	-	-
PC5. perform basic diagnostic tests to confirm that the power source is providing requisite inputs to the Ceiling Fan / Geyser & there are no fault with the same and there are no external faults like improper mounting etc.	1	3	-	-
PC6. isolate the Ceiling Fan / Geyser / Geyser from the power source & disassemble the component(s) or module(s) from the Ceiling Fan / Geyser and perform the functional test(s) to confirm their working	1	3	-	-
PC7. inform the superior if unable to follow the company guidelines and identify the fault	1	1	-	-
Complete the repairs by correcting the faults by replacing the dysfunctional component/module in the Ceiling Fan / Geyser	**8**	**12**	**-**	**-**
PC8. test the basic electrical & electronic components using multi-meter	1	1	-	-
PC9. check the mains outlet with a multi-meter for correct voltage output	1	1	-	-
PC10. correct the minor external faults like loose connections, improper mounting	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. educate the customer to get the major external faults like faulty or intermittent power	1	1	-	-
PC12. replace the faulty component or module and re assemble the Ceiling Fan / Geyser	1	1	-	-
PC13. take help from assistant/junior technician (if any) & train them about the repair process	1	2	-	-
PC14. follow the companys guidelines in case the replacement component(s) or module(s) are not available or any such scenario	1	2	-	-
PC15. update the supervisor & the customer in the above situation	1	2	-	-
Confirm functionality of the repaired unit and post repair activities	**9**	**10**	**-**	**-**
PC16. switch ON the power supply to the Ceiling Fan / Geyser and carry out the functionality test a per the service manual and confirm that it is working normally	1	1	-	-
PC17. demonstrate and confirm the functionality of the Ceiling Fan / Geyser to the customer	1	2	-	-
PC18. complete documentation procedures to record complaint closure, collect the applicable charges from the customer & provide receipt of the same as per company policies	1	1	-	-
PC19. inform the customer about the repairs carried out, preventive care to be taken by them and the usage tips if any and answer to the queries regarding the Ceiling Fan / Geyser or the faults politely to their satisfaction	1	1	-	-
PC20. offer the recommended accessories & extended warranties/AMC to the customer if available.	1	1	-	-
PC21. clean the repair area of all the debris & dispose them with customers permission	1	1	-	-
PC22. collect the defective component(s) or module(s) & handover them to stores	1	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. thank the customer for giving us a chance to serve them & request them to give a positive feedback	1	1	-	-
PC24. inform the superior about the job closure	1	1	-	-
Interact with & train technician	**2**	**4**	**-**	**-**
PC25. interact with service technicians from time to time in-order to understand problems faced on the field	1	2	-	-
PC26. educate junior level technicians about the commonly occurring problems,diagnosis procedures & customer handling	1	2	-	-
Achieve productivity & quality as per companys norms	**10**	**14**	**-**	**-**
PC27. diagnose the problem accurately and in short time	1	2	-	-
PC28. identify the problem modules such as the motor, bush, bearing etc	1	1	-	-
PC29. optimise the time taken to fix the dysfunctional Ceiling Fan / Geyser	1	2	-	-
PC30. rectify to avoid the repeat fault in the Ceiling Fan / Geyser	1	2	-	-
PC31. record minimum customer complaints post service	1	1	-	-
PC32. select the right spares as per recorded complaints at the customer care	1	1	-	-
PC33. educate customer on Ceiling Fan / Geyser maintenance to avoid problems in the future	1	2	-	-
PC34. ensure damage free handling of the Ceiling Fan / Geyser	1	1	-	-
PC35. achieve 100% customer satisfaction	1	1	-	-
PC36. make sale of related products such as new equipment or AMC as applicable	1	1	-	-
Ensure safety at workplace	**5**	**8**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC37. comply with health and safety, environmental and other relevant regulations and organizational guidelines at work	1	2	-	-
PC38. adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while performing the repair operations	1	1	-	-
PC39. ensure work area is clean and safe from hazards	1	1	-	-
PC40. ensure that all tools, equipment, power tool cables, extensions are in a safe and usable condition	1	1	-	-
PC41. dispose of waste items in a safe and environmentally acceptable manner	-	2	-	-
PC42. leave the work area in a safe condition and free from foreign object debris	1	1	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N3124
NOS Name	Repair small home appliances (Geyser & Fan)
Sector	Electronics
Sub-Sector	Consumer Electronics & IT Hardware
Occupation	After Sales Service
NSQF Level	4
Credits	TBD
Version	1.0
Next Review Date	19/04/2018

ELE/N9302: Diagnose and repair fault in LED Light

Description

This OS unit is about diagnosing the fault in the non-functional LED Light and mending it to make the light operational again.

Scope

This unit/ task covers the following:

- Find and repair component-level fault
- Find and repair LED strip-level fault
- Achieve productivity and quality standards

Elements and Performance Criteria

Finding and repairing component level fault

To be competent, the user/individual on the job must be able to:

PC1. connect the non-functional LED Light with the AC source and switch it on

PC2. check that there is no loose, de-soldered wires and connections if the light does not switch on

PC3. solder wires and make connections in case of loose, de-soldered wires and connections to make the light operational again

PC4. dismantle the LED light if no loose, de-soldered wires and connections are found externally

PC5. check the LED light engine with DC supply as per the voltage / current requirements of the product

PC6. replace the LED light engine if it is found faulty

PC7. check the supply unit with AC supply / multimeter to find out the voltage / current output in case LED light Engine is not found defective

PC8. check voltage / current output at different sections of the supply unit with multimeter to find out its damaged section in case of no voltage / current output found in supply unit

PC9. check the components with multimeter individually of the section where voltage output is found to be less than desired / no output

PC10. repair / replace the damaged components / SMPs

PC11. check output voltage/current of the supply unit again with multimeter

PC12. reassemble the LED light if repaired / replaced supply unit is found okay

Finding and repairing LED strip level fault

To be competent, the user/individual on the job must be able to:

PC13. connect the non-functional LED Light with the AC source and switch it on

PC14. check how many LED strips are non-functional / damaged from the array of LED strips in the light

PC15. remove the glass shell from the LED light

PC16. replace the burnt out / damaged LED strips

PC17. check the LED array after connecting it with AC source and switching it on

PC18. replace the glass shell on the LED Light and close it if all the strips are foundoperational

Achieving productivity and quality standards

To be competent, the user/individual on the job must be able to:

PC19. correctly find the root cause of non-functional LED light and repair it in

PC20. document the fault diagnosis and repair process as per SOP

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. companys policies on: incentives, testing & repairing standards and personnel management

KU2. companys standard operating procedures and processes related to LED Luminary product testing and repair

KU3. importance of the individuals role in the workflow

KU4. reporting structure

KU5. safety and quality standards followed in the organization

KU6. various electronic & electrical components, materials and their specific properties & usages

KU7. basics of power electronics and its usages in lighting controls, or LED power supplies and LED drivers

KU8. special safety and handling precautions to be taken during LED luminary testing

KU9. 5S standards (sorting, setting, standardise, sustain, shining) + safety, security

KU10. how to use multimeter, tester, LCR meter and power analyzer

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. complete forms such as work orders, invoices, maintenance records

GS2. note problems on job sheet and details of work done

GS3. read warnings, instructions and other text material on product labels, components, etc.

GS4. read job sheets or work orders

GS5. read product and module serial numbers and interpret details such as make, date, availability

GS6. receive and ask for clarifications from supervisor on the job requirement

GS7. listen carefully to customer and interpret customers statement of symptoms

GS8. communicate in local language

GS9. educate and inform customer about product, contractual issues such as warranty, cost of service and module replacement

GS10. educate on precautions to be taken in order to avoid recurrence of problem

GS11. follow standard operating procedures while making decisions

GS12. take approval from supervisor in case the decision has to be made for exceptions

GS13. work with supervisor and co-workers to achieve smooth workflow

GS14. work with superiors and co-workers to share knowledge and learning

GS15. maintain personal grooming

GS16. be polite, patient and courteous under all circumstances with all types of customers

GS17. decide on the spot on whether interaction of customer with superior is necessary or not

GS18. maintain proper etiquette including the appropriate physical distance with customer during conversation, not entering bedroom without permission

GS19. put customer at ease and generate customers confidence

GS20. seek inputs at assess the problems

GS21. interpret accurately drawings, wiring and job specifications/instructions

GS22. improve work processes

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Finding and repairing component level fault	**21**	**33**	**-**	**-**
PC1. connect the non-functional LED Light with the AC source and switch it on	3	5	-	-
PC2. check that there is no loose, de-soldered wires and connections if the light does not switch on	3	5	-	-
PC3. solder wires and make connections in case of loose, de-soldered wires and connections to make the light operational again	3	3	-	-
PC4. dismantle the LED light if no loose, de-soldered wires and connections are found externally	3	5	-	-
PC5. check the LED light engine with DC supply as per the voltage / current requirements of the product	2	2	-	-
PC6. replace the LED light engine if it is found faulty	1	2	-	-
PC7. check the supply unit with AC supply / multimeter to find out the voltage / current output in case LED light Engine is not found defective	1	2	-	-
PC8. check voltage / current output at different sections of the supply unit with multimeter to find out its damaged section in case of no voltage / current output found in supply unit	1	1	-	-
PC9. check the components with multimeter individually of the section where voltage output is found to be less than desired / no output	1	2	-	-
PC10. repair / replace the damaged components / SMPs	1	2	-	-
PC11. check output voltage/current of the supply unit again with multimeter	1	2	-	-
PC12. reassemble the LED light if repaired / replaced supply unit is found okay	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Finding and repairing LED strip level fault	**13**	**17**	**-**	**-**
PC13. connect the non-functional LED Light with the AC source and switch it on	2	3	-	-
PC14. check how many LED strips are non-functional / damaged from the array of LED strips in the light	3	2	-	-
PC15. remove the glass shell from the LED light	2	3	-	-
PC16. replace the burnt out / damaged LED strips	2	3	-	-
PC17. check the LED array after connecting it with AC source and switching it on	2	3	-	-
PC18. replace the glass shell on the LED Light and close it if all the strips are foundoperational	2	3	-	-
Achieving productivity and quality standards	**6**	**10**	**-**	**-**
PC19. correctly find the root cause of non-functional LED light and repair it in	3	5	-	-
PC20. document the fault diagnosis and repair process as per SOP	3	5	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N9302
NOS Name	Diagnose and repair fault in LED Light
Sector	Electronics
Sub-Sector	Consumer Electronics & IT Hardware
Occupation	Sales And After Sales Service
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/06/2015
Next Review Date	31/03/2022
NSQC Clearance Date	NA

ELE/N9919: Work with superiors and colleagues

Description

This OS unit is about communicating, coordinating and maintaining proper relationship with colleagues and seniors in order to achieve smooth work flow

Scope

This unit/ task covers the following:

- Interact with supervisor or superior
- Coordinate with colleagues

Elements and Performance Criteria

Interacting with supervisor

To be competent, the user/individual on the job must be able to:

PC1. understand work requirements by receiving instructions from reporting supervisor

PC2. understand standard operating procedure of the company

PC3. escalate problems that cannot be handled including repetitive PCB defects, machine failures, potential hazards, process disruptions, repairs and maintenance of machine

PC4. report work completed and receive feedback on work done

PC5. resolve personnel issues

PC6. rectify errors as per feedback and minimize mistakes to zero in future

PC7. communicate about process flow improvements, quality of output, product defects received from previous process, repairs and maintenance of tools and machinery as required and find technical solutions on specific issues

PC8. handover completed work and deliver the work of expected quality despite constraints

Interacting with colleagues

To be competent, the user/individual on the job must be able to:

PC9. collect required spares and raw materials from tool room or stores

PC10. deposit unused or faulty materials, parts and tools to stores

PC11. assist colleagues where necessary and as per capability

PC12. resolve conflicts with colleagues at work to achieve smooth workflow

PC13. complete rework in time based on feedback from quality or process departments

PC14. put team over individual goals

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. companys policies on: incentives, delivery standards, and personnel management

KU2. work flow involved in companys process

KU3. importance of the individuals role in the workflow

KU4. reporting structure
KU5. how to communicate effectively
KU6. how to build team coordination

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. deliver product to next work process on time
GS2. to share work load as required
GS3. how to report potential areas of disruptions to work process
GS4. when to report to supervisor and when to deal with a colleague depending on the type of concern
GS5. to reduce repetitive errors and improve work process
GS6. how to spot process disruptions and delays

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Interacting with supervisor	**20**	**32**	**-**	**-**
PC1. understand work requirements by receiving instructions from reporting supervisor	2	4	-	-
PC2. understand standard operating procedure of the company	2	4	-	-
PC3. escalate problems that cannot be handled including repetitive PCB defects, machine failures, potential hazards, process disruptions, repairs and maintenance of machine	2	4	-	-
PC4. report work completed and receive feedback on work done	2	4	-	-
PC5. resolve personnel issues	3	4	-	-
PC6. rectify errors as per feedback and minimize mistakes to zero in future	3	4	-	-
PC7. communicate about process flow improvements, quality of output, product defects received from previous process, repairs and maintenance of tools and machinery as required and find technical solutions on specific issues	3	4	-	-
PC8. handover completed work and deliver the work of expected quality despite constraints	3	4	-	-
Interacting with colleagues	**20**	**28**	**-**	**-**
PC9. collect required spares and raw materials from tool room or stores	3	5	-	-
PC10. deposit unused or faulty materials, parts and tools to stores	3	5	-	-
PC11. assist colleagues where necessary and as per capability	3	5	-	-
PC12. resolve conflicts with colleagues at work to achieve smooth workflow	3	5	-	-
PC13. complete rework in time based on feedback from quality or process departments	4	4	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. put team over individual goals	4	4	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N9919
NOS Name	Work with superiors and colleagues
Sector	Electronics
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/03/2014
Next Review Date	31/03/2022
NSQC Clearance Date	NA

ELE/N9921: Follow safety standards

Description

This OS unit is about following safety procedures, communicating potential hazards and dangers of accidents on the job

Scope

This unit/ task covers the following:

- Understand potential sources of accidents
- Use safety gear to avoid accidents
- Understand the safety procedures followed by the company
- Follow daily safety measure
- Communicate to supervisor

Elements and Performance Criteria

Understanding potential sources of accidents

To be competent, the user/individual on the job must be able to:

PC1. spot and report potential hazards on time

PC2. follow company policy and rules regarding hazardous materials

PC3. avoid accidents related to use of potentially dangerous chemicals, gases, sharp tools and hazards from machines which involves exposure to possible injuries such as cuts, bites, stings, minor burns, etc.

PC4. handle with care when using an electrical drill and sharp cutting objects

Using safety gear

To be competent, the user/individual on the job must be able to:

PC5. identify which safety gear must be used for a particular task

PC6. eye, respiratory and hearing protection as per company policy

PC7. use safety gear such as respirator, mask, skull caps, gloves, googles, jacket , etc., as prescribed for the job

Understanding of safety procedures

To be competent, the user/individual on the job must be able to:

PC8. comply with standard health and safety procedure followed in the company while handling an equipment and hazardous materials and tools or situations

PC9. identify and follow the evacuation procedure properly such as fire drills, emergency evacuation procedures, first aid to self and others, etc., which help in case of an emergency

Following daily safety measure

To be competent, the user/individual on the job must be able to:

PC10. take adequate safety measures while on work to prevent accidents

PC11. ensure zero accidents in work

PC12. avoid damage of components due to negligence in ESD procedures

PC13. ensure no loss for company due to safety negligence

PC14. ensure proper machine maintenance, work process achieving quality outputs as per the company standard

Communicating to supervisor

To be competent, the user/individual on the job must be able to:

PC15. improve process flow to reduce anticipated or repetitive hazards

PC16. report on mishandling of tools, machines or hazardous materials and on electrical problems that could result in accident

PC17. escalate about any hazardous materials or things found in the premises

PC18. report about any breach of safety procedure in the company

PC19. follow electrostatic discharge (ESD) measures for electronic component safety

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. companys policies on handling: harmful chemicals and sharp tools, safety and hazards of machines, fire safety/drill, first aid and, disposal of harmful chemicals and materials, quality standards

KU2. company occupational safety and health policy followed

KU3. company emergency evacuation procedure

KU4. companys medical policy

KU5. how to maintain the work area safe and secure

KU6. how to handle hazardous material

KU7. how to follow safety procedures while operating hazardous tools and equipment

KU8. emergency procedures to be followed such as fire accidents and fire safety education

KU9. how to use machines and tools without causing bodily harm

KU10. first aid execution

KU11. disposal of hazardous chemicals, tools and materials by following prescribed environmental norms or as per company policy

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. to effectively communicate the danger

GS2. to understand the quality standard of the company

GS3. to learn from past mistakes regarding use of hazardous machines, tools or chemicals

GS4. how to spot danger

GS5. procedure to follow in the event of a fire or other hazard

GS6. to wear gloves, goggles, masks, caps, shoes, coats, etc

GS7. to use safety equipment such as fire extinguisher during fire accidents

GS8. importance of reporting potential sources of danger

GS9. appropriate actions to be taken in the event of an accident

GS10. procedure for disposing of hazardous materials, safely and following environmental guidelines

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Understanding potential sources of accidents	**8**	**12**	**-**	**-**
PC1. spot and report potential hazards on time	2	3	-	-
PC2. follow company policy and rules regarding hazardous materials	2	3	-	-
PC3. avoid accidents related to use of potentially dangerous chemicals, gases, sharp tools and hazards from machines which involves exposure to possible injuries such as cuts, bites, stings, minor burns, etc.	2	3	-	-
PC4. handle with care when using an electrical drill and sharp cutting objects	2	3	-	-
Using safety gear	**9**	**11**	**-**	**-**
PC5. identify which safety gear must be used for a particular task	3	3	-	-
PC6. eye, respiratory and hearing protection as per company policy	3	4	-	-
PC7. use safety gear such as respirator, mask, skull caps, gloves, googles, jacket , etc., as prescribed for the job	3	4	-	-
Understanding of safety procedures	**8**	**12**	**-**	**-**
PC8. comply with standard health and safety procedure followed in the company while handling an equipment and hazardous materials and tools or situations	4	6	-	-
PC9. identify and follow the evacuation procedure properly such as fire drills, emergency evacuation procedures, first aid to self and others, etc., which help in case of an emergency	4	6	-	-
Following daily safety measure	**10**	**10**	**-**	**-**
PC10. take adequate safety measures while on work to prevent accidents	2	2	-	-
PC11. ensure zero accidents in work	2	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. avoid damage of components due to negligence in ESD procedures	2	2	-	-
PC13. ensure no loss for company due to safety negligence	2	2	-	-
PC14. ensure proper machine maintenance, work process achieving quality outputs as per the company standard	2	2	-	-
Communicating to supervisor	**5**	**15**	**-**	**-**
PC15. improve process flow to reduce anticipated or repetitive hazards	1	3	-	-
PC16. report on mishandling of tools, machines or hazardous materials and on electrical problems that could result in accident	1	3	-	-
PC17. escalate about any hazardous materials or things found in the premises	1	3	-	-
PC18. report about any breach of safety procedure in the company	1	3	-	-
PC19. follow electrostatic discharge (ESD) measures for electronic component safety	1	3	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N9921
NOS Name	Follow safety standards
Sector	Electronics
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/03/2014
Next Review Date	31/03/2022
NSQC Clearance Date	NA

FIC/N9005: Evaluate and develop entrepreneur skills

Description

This OS unit is about evaluating and developing entrepreneur skills before starting a food processing unit

Elements and Performance Criteria

Evaluate before starting foodprocessing unit

To be competent, the user/individual on the job must be able to:

PC1. self evaluate on the capability to start usiness, develop business, manage an organization, manage time, handle ifferent people (customers, vendors, government officials, bankers, onsultants, etc),make independent and clear decisions under pressure, physical and emotional stamina work long hours

PC2. evaluate the performance of various food processing sectors and sale/market share of various category of processed foods, to decide on starting the food processing sector and food product

PC3. choose the right product based on trengths,potential,capability, market demand,profitability,personal preferences

PC4. conduct market survey to understand the market trend,market needs, opportunity, competition

PC5. review market demand based on ompetitors,customers, market requirement, current market status etc

PC6. consult with experts, experienced people and family on the ideas developed

Develop Entrepreneur Skills

To be competent, the user/individual on the job must be able to:

PC7. acquire knowledge (through training or other sources like reading books) on communication skills, management skills, accounting skills, marketing skills

PC8. develop / acquire technical skills (through training or through work experience) on raw materials handling product processing, productpreservation, packaging ,quality control, roduct storage, processing machineries, relevant food laws and regulations, food safety hygiene and sanitation

PC9. develop skills on distribution, sales and marketing (through training or discussing and learning from experienced people)

PC10. learn to be realistic and objective while planning business, and discrete in sharing the ideas

PC11. acquire knowledge (through training or other sources like reading books) on communication skills, management skills, accounting skills,marketing skills

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. relevant organisational standards, process standards and procedures required for the food processing unit

KU2. performance evaluation of food processing units

KU3. decision making on products to be produced in the organisation

KU4. methods and importance of market survey
KU5. understanding market demand
KU6. methods and importance of consulting with experts
KU7. various food processing industries, market trend and market share of various processed food
KU8. technical requirement for food processing sector like raw materials, packaging materials, process etc
KU9. food processing machineries
KU10. quality requirement for food
KU11. food laws and regulations
KU12. food safety and hygiene
KU13. good manufacturing practice (GMP)
KU14. hazard analysis and critical control point (HACCP)

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. note thet information to be communicated
GS2. fill relevant applications required for food processing units
GS3. note the information required for establishing and operating food processing unit
GS4. document the process, process equipments and parameters for products processed
GS5. record the raw materials, finished products produced, inventory, stock distribution, marketing and sales
GS6. note down observations (if any) related to the process or organisation
GS7. write communications to government officials, financial institutions and employees
GS8. note down the data for erp or as required by the organization
GS9. read communications from various government departments
GS10. read and interpret and process flowchart and process required for all products produced
GS11. read internal communications from the employees
GS12. read communications from market, various trade related organisations
GS13. discuss task lists, schedules and activities with the employees
GS14. effectively communicate with the employees
GS15. question the employees in order to understand the nature of the problem and to clarify queries
GS16. attentively listen and comprehend the information given by the speaker
GS17. communicate clearly with the employees to understand and resolve issues
GS18. communicate clearly with the vendors, government officials, bankers, employees, customers, consumers etc with respect to organisation, process, product, sales etc
GS19. analyse critical points in day to day tasks through experience and observation and identify control measures to solve the issue
GS20. handle and resolve issues related to entire operation, in case of issues beyond the capability of the employees
GS21. plan and organize the work

GS22. plan and allot work/responsibilities to the employees

GS23. organize raw materials and packaging materials required for all products produced in the organisation

GS24. plan to prioritize work based on organisational needs

GS25. plan to prioritize the work based on the order/market requirement

GS26. plan to utilize the time and equipments effectively

GS27. plan to utilise the time effectively

GS28. support the employees in their tasks to achieve production and sales

GS29. understand customer requirements and their priority and respond as per their needs

GS30. support employees in solving problems by understanding the problems

GS31. arrive at possible solution for problems related to operation, by discussing with experienced/concerned people

GS32. apply domain information about maintenance processes and technical knowledge about tools and equipment

GS33. use common sense and make judgments on day to day basis

GS34. use reasoning skills to identify and resolve basic problems

GS35. use intuition to detect any potential problems which could arise during operations

GS36. use acquired knowledge of the process for identifying and handling issues

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Evaluate before starting foodprocessing unit	**19**	**41**	**-**	**-**
PC1. self evaluate on the capability to start usiness, develop business, manage an organization, manage time, handle ifferent people (customers, vendors, government officials, bankers, onsultants, etc),make independent and clear decisions under pressure, physical and emotional stamina work long hours	5	10	-	-
PC2. evaluate the performance of various food processing sectors and sale/market share of various category of processed foods, to decide on starting the food processing sector and food product	3	7	-	-
PC3. choose the right product based on trengths,potential,capability, market demand,profitability,personal preferences	3	7	-	-
PC4. conduct market survey to understand the market trend,market needs, opportunity, competition	3	7	-	-
PC5. review market demand based on ompetitors,customers, market requirement, current market status etc	3	7	-	-
PC6. consult with experts, experienced people and family on the ideas developed	2	3	-	-
Develop Entrepreneur Skills	**16**	**24**	**-**	**-**
PC7. acquire knowledge (through training or other sources like reading books) on communication skills, management skills, accounting skills, marketing skills	4	6	-	-
PC8. develop / acquire technical skills (through training or through work experience) on raw materials handling product processing, productpreservation, packaging ,quality control, roduct storage, processing machineries, relevant food laws and regulations, food safety hygiene and sanitation	4	6	-	-
PC9. develop skills on distribution, sales and marketing (through training or discussing and learning from experienced people)	4	6	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. learn to be realistic and objective while planning business, and discrete in sharing the ideas	2	3	-	-
PC11. acquire knowledge (through training or other sources like reading books) on communication skills, management skills, accounting skills,marketing skills	2	3	-	-
NOS Total	**35**	**65**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	FIC/N9005
NOS Name	Evaluate and develop entrepreneur skills
Sector	Food Processing
Sub-Sector	Fruits and Vegetables, Food Grain Milling, Dairy Products, Meat and Poultry, Fish and Sea Food, Bread and Bakery, Alcoholic Beverages, Aerated Water/Soft Drinks, Soya Food, Packaged Foods
Occupation	Processing-Bread and Bakery
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	30/09/2021
Next Review Date	29/06/2024
NSQC Clearance Date	30/09/2021

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N3101.Engage with customer for service	40	60	-	-	100	15
ELE/N3118.Install the water purifier	40	60	-	-	100	10
ELE/N3119.Repair dysfunctional water purifier	40	60	-	-	100	10
ELE/N3120.Repair dysfunctional mixer/juicer/grinder	40	60	-	-	100	10
ELE/N3124.Repair small home appliances (Geyser & Fan)	40	60	-	-	100	10
ELE/N9302.Diagnose and repair fault in LED Light	40	60	-	-	100	15
ELE/N9919.Work with superiors and colleagues	40	60	-	-	100	10
ELE/N9921.Follow safety standards	40	60	-	-	100	10
FIC/N9005.Evaluate and develop entrepreneur skills	35	65	-	-	100	10
Total	**355**	**545**	**-**	**-**	**900**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.